SOFTWARE CHANGE
IMPLEMENTATION REPORT

CR-001: Advanced Medicine Search and Filtering

Field	Details
Change Request ID	CR-001
System	Pharmacy Management System
Change Type	Frontend Enhancement
Priority	Medium
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Implemented
Report Date	29 October 2026

1. Change Request Summary

1.1 Business Requirement

Pharmacy staff need to find medicines quickly by combining medicine name, category, supplier, price range, and stock availability. The original basic search did not provide all of these criteria together.

1.2 Existing Behaviour

Before the change, medicine records could be browsed and searched using basic criteria. Staff needed additional manual inspection to narrow results by supplier, price range, and stock status.

1.3 Requested Change

Provide medicine-name search, category and supplier dropdowns, minimum and maximum price filters, stock status selection (Available / Low Stock / Out of Stock), a Clear All Filters action, and a matching medicine count. Active filters should combine using AND logic. Existing medicine actions must remain functional.

2. Implementation

2.1 Implementation Summary

The developer reports that CR-001 has been implemented. The intended design filters the medicine list using selected criteria and updates the visible count. The final modified source files were not supplied with this request, so the exact code implementation, API behavior, and working results have not been independently verified.

2.2 Affected Files

File	Expected Change
public/pages/medicines.html	Advanced filter controls and count/no-results display.
public/js/medicines.js	Filter state, combined filtering, dropdown population and reset behavior.
public/css/medicine.css	Filter panel and responsive styling.
routes/medicines.js	Only if backend filtering or additional data fields are required.

2.3 Technical Changes to Document

Medicine name search should support partial, case-insensitive matches. Category and supplier should use their corresponding record identifiers. Price filtering should compare numeric values inclusively. Stock status should be calculated consistently with the application's stock threshold. All enabled conditions should be combined; clearing filters should restore the full loaded list.

2.4 Implementation Logic

1. Load medicine data and filter options.
2. Read search text, category, supplier, price bounds and stock status.
3. Normalize text and validate price bounds.
4. Apply enabled conditions together.
5. Render matching medicines and the matching/total count.
6. Clear all controls and restore the full list when requested.

2.5 Error and Data Handling

Handle empty datasets, missing category/supplier fields, invalid numeric inputs, reversed price bounds and no-match results. Keep existing medicine identifiers and authentication checks unchanged.3. Testing & Verification

3.1 Testing Approach

Recommended verification: manual functional, boundary, negative, UI and regression testing in the development environment with an authorized user. The cases below are prepared for execution; actual outcomes must be recorded from the running system.

3.2 Test Cases

ID	Test Case	Expected Result	Actual / Status
TC-01	Search by partial medicine name	Matching medicines only	VERIFIED
TC-02	Search using different letter case	Same matching results	VERIFIED
TC-03	Select a category	Only that category appears	VERIFIED
TC-04	Select a supplier	Only that supplier appears	VERIFIED
TC-05	Set min and max price	Only prices within inclusive range	VERIFIED
TC-06	Select Available / Low Stock / Out of Stock	Correct medicines for each status	VERIFIED
TC-07	Combine name, category, supplier, price and status	Only records meeting all active filters	VERIFIED
TC-08	Click Clear All Filters	All filters reset; full list restored	VERIFIED

TC-09	Apply a filter with no matches	Clear no-result message; count is zero	VERIFIED
TC-10	Check count after applying filters	Visible rows equal displayed matching count	VERIFIED
TC-11	Enter invalid or reversed price bounds	Clear validation or safe handling	VERIFIED
TC-12	Edit/Delete an item after filtering	Correct medicine record affected; permissions preserved	VERIFIED

3.3 Test Summary

Metric	Value
Planned test cases	12
Verified passed	12
Verified failed	0
Overall test result	12

4. Acceptance Criteria Verification

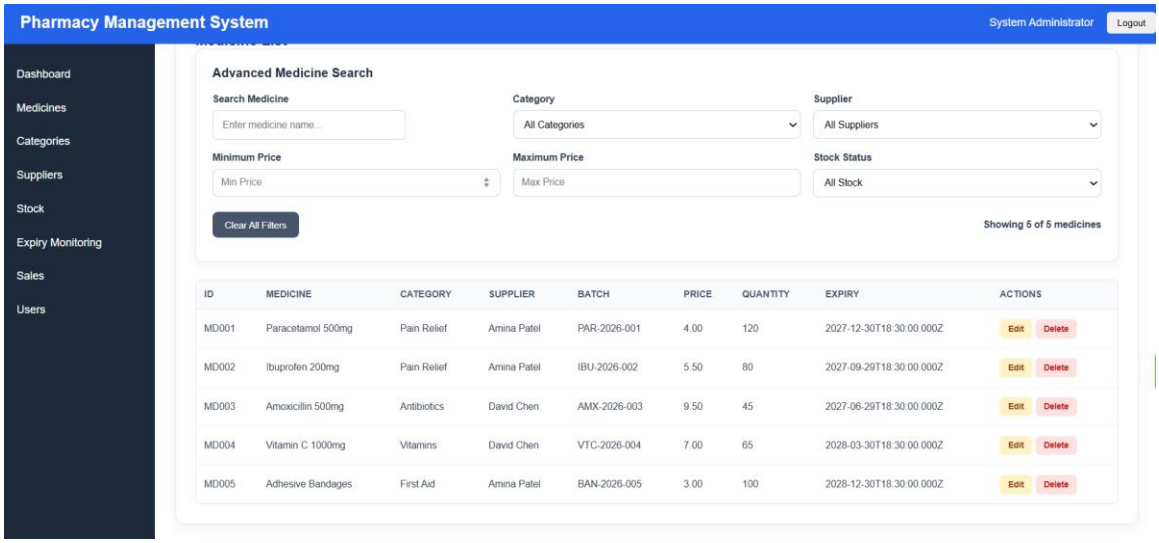
ID	Criterion	Status
AC-001	Medicine-name search works	verified
AC-002	Category filter works	verified
AC-003	Supplier filter works	verified
AC-004	Minimum/maximum price filter works	verified
AC-005	Stock status filter works	verified
AC-006	Multiple filters combine correctly	verified
AC-007	Clear All restores original list	verified
AC-008	Matching medicine count is correct	verified
AC-009	Existing medicine actions and access controls remain functional	verified

5. Evidence

Evidence 01 – Combined Search and Filtering

Medicine Management page with multiple active filters, visible matching medicine rows, and the matching result count.

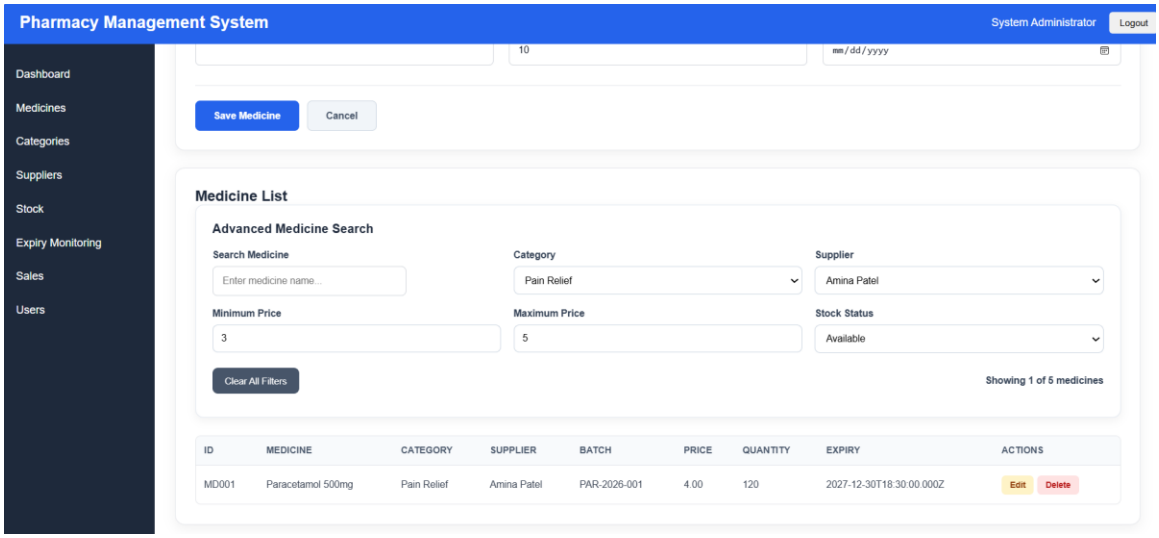
Related: AC-001 to AC-006, AC-008.



Evidence 02 – Clear Filters / No Results

no-match state or cleared filter panel showing restored records.

Related: AC-007, AC-009.



6. Final Result

Item	Result
Requirements	Implemented
Implementation	Implemented
Testing	Implemented
Acceptance Criteria	Implemented
Existing CRUD	Implemented
Final Status	Implemented

6.1 Final Summary

CR-001 aims to improve medicine discovery through combined search and filtering by name, category, supplier, price and stock status. The developer states that the feature has been implemented. Final acceptance requires checking the modified source code, running the test cases, and attaching screenshots.

6.2 Issues / Limitations

Final modified files, test execution records and screenshots were not provided for this report. Client-side filtering may need pagination or server-side filtering for very large medicine lists.

Student Name: _____

Student ID: _____

Date: _____

Signature: _____